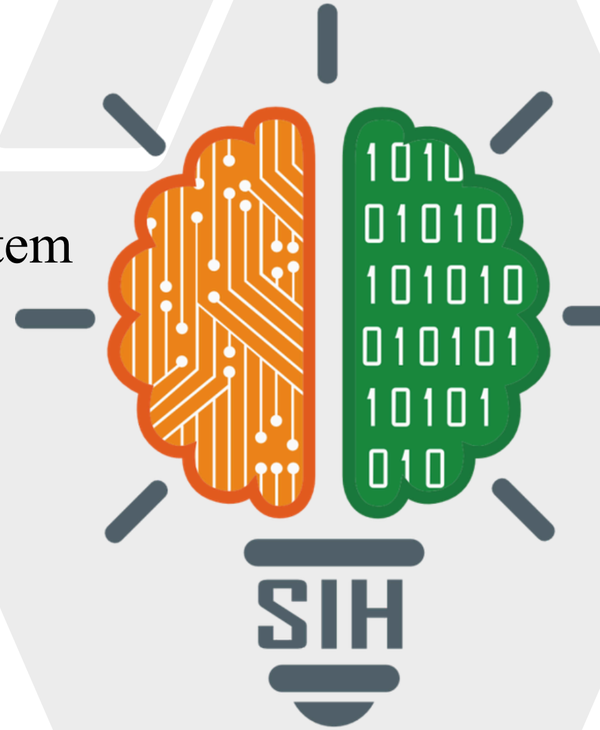
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TITLE PAGE

- **Problem Statement ID** - SIH 26262
- **Problem Statement Title -** Integrated AICTE Approval Workflow Automation Compliance Monitoring and Alert Management System
- **Theme** - Smart Automation
- **PS Category** - Software
- **Team ID** - 199751
- **Team Name** - Phoenix Crew



Phoenix Crew

WAYMARK - THE WORKFLOW AUTOMATER

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The Problem

- AICTE approval processes involve stages many officers, many document requirements and many time-bound obligations.
- Manual follow-ups and disconnected spreadsheets make it hard to see how many AICTE applications are still pending, which AICTE documents are about to expire and whether AICTE compliance is being met.
- Complaints, about the institution may go to different bureaus so that there is no single view of the AICTE case.

Impact

- Delays in getting approvals because the stage-level issues were not noticed.
- Missing compliance. Documents that had expired.
- Handling the complaint multiple times across different bureaus.
- It was not clear who was responsible and there was not accountability, within the institution.
- There was no view of the institutions regulatory history.

Our Solution

- **Automates Tracking:** WAYMARK watches over approval steps, who owns tasks and the deadlines set by service level agreements.
- **Prevents Missed Deadlines:** WAYMARK keeps an eye on how long documents are valid when EVC meetings happen and what compliance duties exist sending early warnings.
- **Connects Complaints:** WAYMARK gives a view across all bureaus of complaints and finds duplicates.
- **Enables Oversight:** WAYMARK brings together applications, complaints, documents and duties into a single timeline for the institution.
- **Preserves Human Authority:** WAYMARK offers explanations, to officers and keeps the final regulatory choices in the hands of authorised people.

"A delay or missed obligation at one stage can affect the entire approval lifecycle."

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TECHNICAL APPROACH

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Architecture: Modular monolith, AI-assisted regulatory workflow platform.

Frontend: Next.js – Role-based dashboards and application tracking.

Backend: FastAPI – Workflow processing, authentication and business logic.

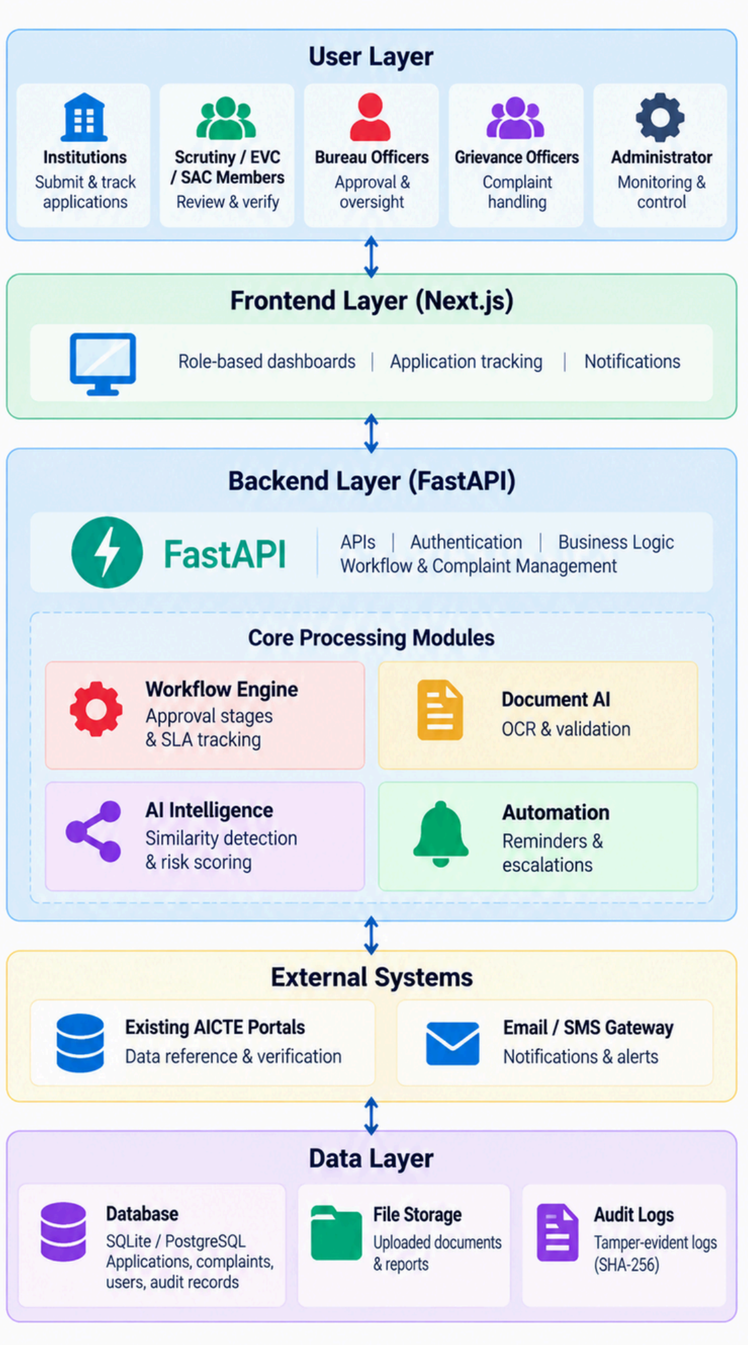
Database: SQLite / PostgreSQL – Applications, complaints, users and audit records.

Document AI: OCR-based document extraction and validation.

AI Intelligence: Complaint similarity detection, delay prediction and risk scoring.

Automation: SLA monitoring, deadline reminders and escalation alerts.

Security: JWT authentication, role-based access and tamper-evident audit logs.



What makes WAYMARK different?

- Cross-Bureau Complaint Intelligence
- Proactive Regulatory Monitoring
- Unified Institutional History

"From tracking individual applications to understanding the complete regulatory journey of an institution."

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FEASIBILITY AND VIABILITY

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1. Technical Feasibility

- Built using open-source technologies (Next.js, FastAPI, SQL).
- Modular architecture supports independent feature development.
- OCR, TF-IDF and rule-based analytics enable intelligent processing.

2. Operational Feasibility

- Role-based dashboards simplify officer and institutional operations.
- Automated reminders reduce manual follow-ups.
- Unified institutional records improve coordination across bureaus.

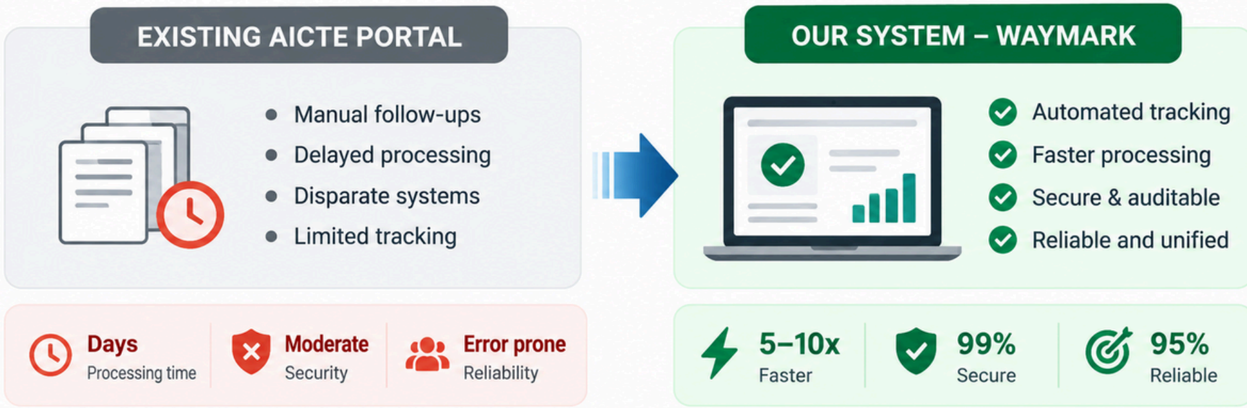
3. Economic Viability

- Open-source stack reduces licensing costs.
- Cloud deployment enables cost-effective infrastructure.
- Modular design reduces future maintenance expenses.

4. Scalability & Sustainability

- PostgreSQL supports growing institutional records.
- Modular services allow future integrations with AICTE portals.
- Configurable workflows adapt to changing regulatory requirements.

Existing AICTE Portal	Our Portal – WAYMARK
Application-centric services	Institution-centric management
Basic application status tracking	Stage-wise tracking with officer ownership
Manual follow-ups for pending activities	Automated SLA alerts and escalations
Separate document submission and verification	OCR-assisted document extraction and expiry tracking
Complaint handling through existing channels	Unified cross-bureau complaint tracking and duplicate
Portal-specific reports	Consolidated analytics, institutional risk and bottleneck



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IMPACT AND BENEFITS

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1. Faster Workflow Management

- This helps cut down on the need to manually check in on things by using reminders.
- It makes it easier to see which steps still need approval.

2. Improved Accountability

- It gives people the job of handling each task.
- It keeps a record of everything that happens.

3. Proactive Compliance

- It keeps track of when documentsre, about to expire and when rules need to be followed.
- It tells the people responsible before they miss a deadline.

4. Unified Complaint Resolution

- Unified Complaint Resolution identifies complaints that might be duplicates across bureaus.
- Unified Complaint Resolution improves how complaints are categorised and tracked.

5. Data-Driven Decisions

- Data-Driven Decisions highlights where workflow slows down and shows risk indicators for the institution.
- Data-Driven Decisions provides analytics that administrators can use.

6. Better Institutional Transparency

- Better Institutional Transparency creates a timeline, for the institution, improves visibility into approvals, compliance and complaints.

Expected Beneficiaries

AICTE Officers

Better workflow visibility and reduced manual work

EVC / SAC Members

Organised inspection and verification records

Bureau Heads

Cross-bureau complaint visibility and oversight

Institutions

Transparent application and compliance tracking

Administrators

Consolidated analytics for better decision making

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RESEARCH AND REFERENCES

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1. AI-Based Intelligent Document Management System (2023)

Procedia Computer Science, Vol. 230, pp. 725–736

Relevance: AI-based document processing and information extraction. Research Paper

2. AI-Powered Intelligent Document Management System (2024)

Information, Vol. 15, Issue 8, Article 461

Relevance: Automated document classification and administrative document management. Research Paper

3. Unveiling the Causes of Waiting Time in Business Processes (2024)

Relevance: Workflow delay analysis and bottleneck identification using event logs. Research Paper

4. Lightweight String Similarity Approaches for Duplicate Detection (2025)

Journal of Informatics and Web Engineering, Vol. 4, Issue 3

Relevance: Text similarity techniques for identifying duplicate complaints. Research Paper

5. PM4Py: A Process Mining Library for Python (2023)

Software Impacts, Vol. 17, Article 100556

Relevance: Process analysis and workflow performance monitoring. Research Paper

OFFICIAL REFERENCE

AICTE Approval Process Handbook 2024–2027

Relevance: Regulatory approval stages, inspections and institutional compliance. Official Handbook (PDF)

PROJECT LINKS

GitHub Repository: View Source Code

Prototype Images: View Project Screenshots

Demo Video: Watch Project Demonstration